UNIVERSITY OF THE PUNJAB

Punjab University College of Information Technology

Allama Iqbal Campus

FILE INTEGRITY CHECKER AND PROTECTION SYSTEM

Course: Information Security

Submitted By: Shanza Shafiq

Roll No: BITF24M008

Submitted To: Sir Huzaifa Nazir

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1. Introduction

In modern computer systems, data security is a critical concern. Important files such as documents, reports, and system files can be modified without the user's knowledge. This may lead to data corruption, loss of important information, or security breaches.

To address this issue, this project presents a **File Integrity Checker and Protection System** that uses cryptographic hashing (SHA-256) to ensure that files remain unchanged. The system allows users to monitor selected files and detect any unauthorized modifications.

2. Problem Statement

Many users do not have a simple method to verify whether their files have been altered or tampered with. Unauthorized modifications can occur silently and may cause serious problems such as data loss or compromised system integrity.

Therefore, there is a need for a system that can:

- Detect file changes accurately
- Alert the user about modifications
- Maintain a record of file activity

3. Objectives

The main objectives of this project are:

- To implement a file integrity monitoring system
- To generate unique hash values using SHA-256
- To detect unauthorized file modifications
- To provide a simple and user-friendly interface
- To maintain logs of all activities
- To implement basic password-based security

4. Methodology / Working

The system works in two main phases:

4.1 Baseline Phase

- User selects a file
- System reads the file in binary mode
- SHA-256 hash is generated
- Hash is stored in a database file (hash.txt)

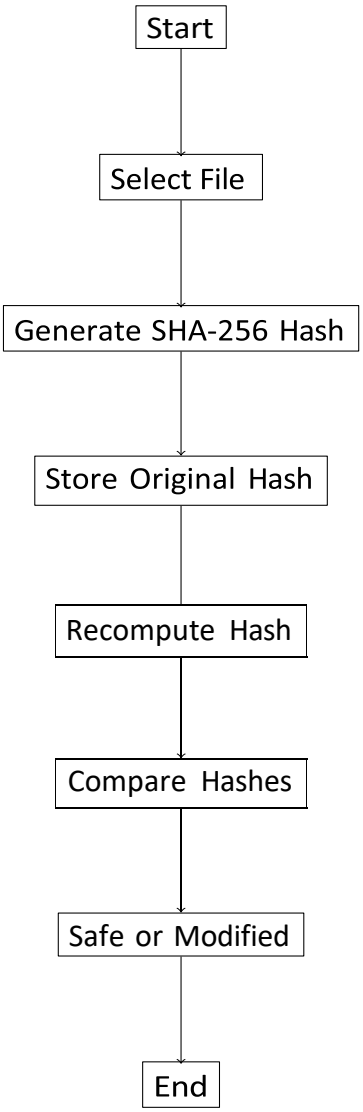
4.2 Verification Phase

- User selects the same file
- System recalculates the hash
- New hash is compared with stored hash

Result:

- If hashes match → File is SAFE
- If hashes differ → File is MODIFIED

5. System Flowchart



6. Implementation Details

6.1 Programming Language

C++

6.2 Libraries Used

- iostream – Input/Output
- fstream – File handling
- string – String operations
- ctime – Date and time

- conio.h – Hidden password input

6.3 Core Components

- SHA-256 Algorithm (manually implemented)
- File reading in binary mode
- Hash storage system (hash.txt)
- Logging system (log.txt)
- Password authentication

6.4 Main Functions

- computeSHA256() – Generates hash
- readFile() – Reads file data
- addFile() – Stores baseline hash
- checkFile() – Verifies integrity
- login() – Password protection

7. Results

The system was successfully implemented and tested. The results show that:

- The system correctly generates SHA-256 hash values
- It detects even the smallest file changes
- It accurately identifies safe, modified, and missing files
- The logging system records all activities with timestamps

8. Conclusion

The project successfully demonstrates how **SHA-256 hashing** can be used to ensure file integrity. It provides a reliable method to detect unauthorized file modifications.

The system is simple, efficient, and useful for local file monitoring. This concept can be further extended into advanced security systems such as real-time monitoring and intrusion detection tools.

9. Future Enhancements

- Real-time file monitoring
- GUI-based interface
- Encryption for file security
- Cloud-based monitoring system